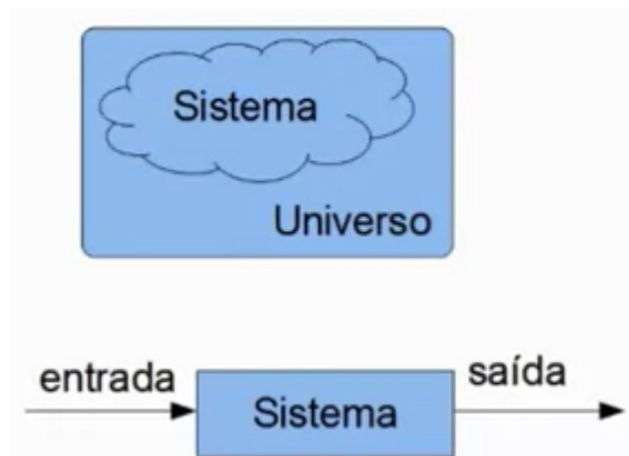


Laboratório de Controle Automático

Função de transferência de malha fechada e as vantagens da realimentação

BIBO estabilidade

estabilidade



$$G(s) = \frac{Y(s)}{U(s)}$$

Transformada de Laplace da entrada

Transformada de Laplace da saída

Estabilidade

↕

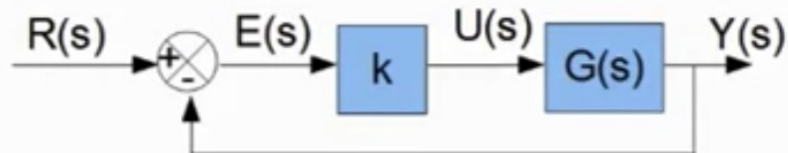
Todos os polos com parte real negativa.

Malha Aberta:



Malha Fechada:



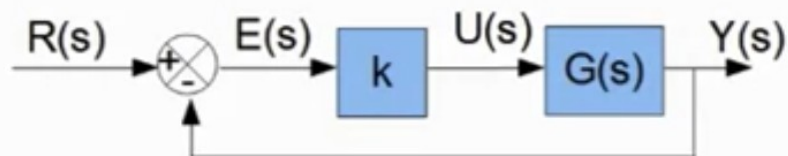


$$G(s) = \frac{Y(s)}{U(s)} \quad \frac{Y(s)}{R(s)} = ?$$

$$U(s) = k E(s) \quad Y(s) = G(s) U(s) \quad Y(s) = G(s) k E(s)$$

$$E(s) = R(s) - Y(s) \quad Y(s) = k G(s) [R(s) - Y(s)]$$

$$Y(s) = k G(s) R(s) - k G(s) Y(s)$$



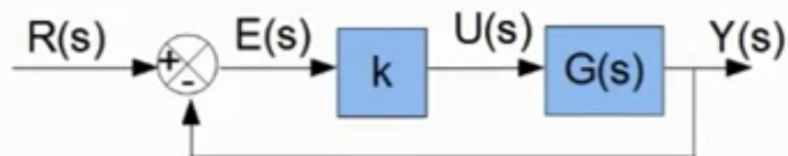
$$G(s) = \frac{Y(s)}{U(s)}$$

$$\frac{Y(s)}{R(s)} = \frac{k G(s)}{1 + k G(s)}$$

$$Y(s) = k G(s) R(s) - k G(s) Y(s)$$

$$Y(s) + k G(s) Y(s) = k G(s) R(s)$$

$$[1 + k G(s)] Y(s) = k G(s) R(s)$$

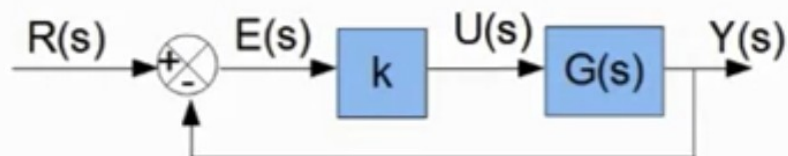


$$G(s) = \frac{Y(s)}{U(s)}$$

$$\frac{Y(s)}{R(s)} = \frac{k G(s)}{1 + k G(s)}$$

$$\frac{Y(s)}{R(s)} = \frac{k \frac{N_G(s)}{D_G(s)}}{1 + k \frac{N_G(s)}{D_G(s)}}$$

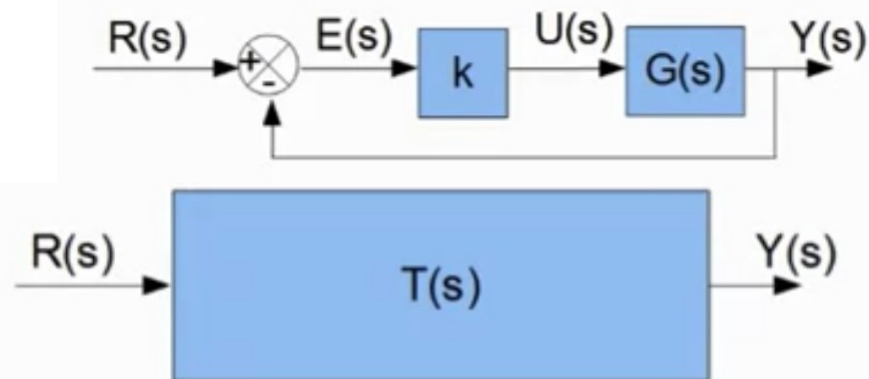
$$= \frac{k N_G(s)}{D_G(s) + k N_G(s)}$$



$$G(s) = \frac{N_G(s)}{D_G(s)}$$

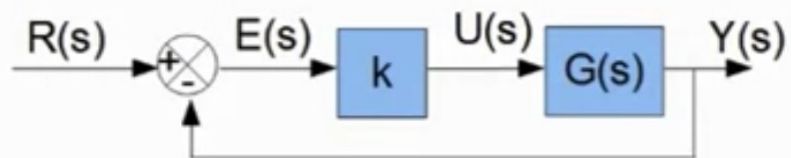
$$T(s) = \frac{k G(s)}{1 + k G(s)}$$

$$T(s) = \frac{k N_G(s)}{D_G(s) + k N_G(s)}$$



$$G(s) = \frac{N_G(s)}{D_G(s)}$$

$$T(s) = \frac{k G(s)}{1 + k G(s)} = \frac{k N_G(s)}{D_G(s) + k N_G(s)}$$

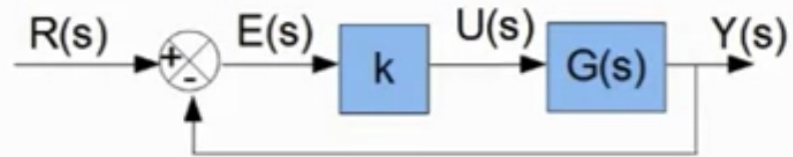


$$G(s) = \frac{1}{s-1}$$

$$T(s) = \frac{k}{s-1+k}$$

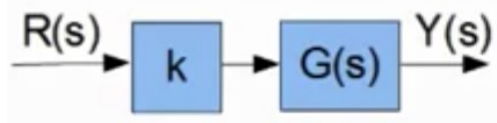
$$k=2 \quad \rightarrow \quad T(s) = \frac{2}{s+1}$$



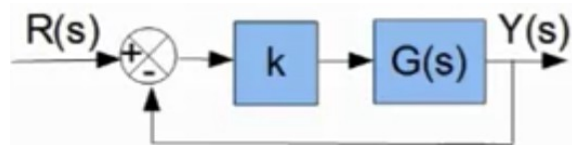


$$G(s) = \frac{1}{s^2 - 2s - 3}$$

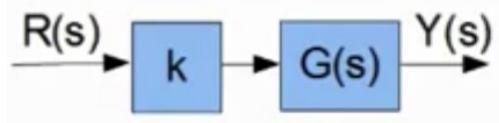
$$T(s) = \frac{k}{s^2 - 2s - 3 + k}$$



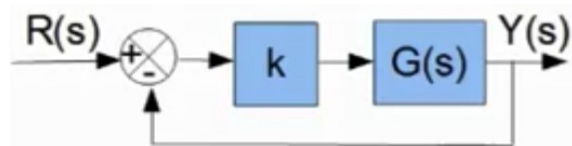
$$Y_N(s) = k A R(s)$$



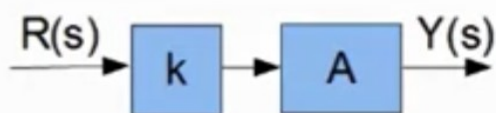
$$Y_N(s) = \frac{k A}{1 + k A} R(s)$$



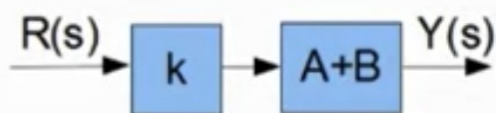
$$Y_R(s) = k(A + B)R(s)$$



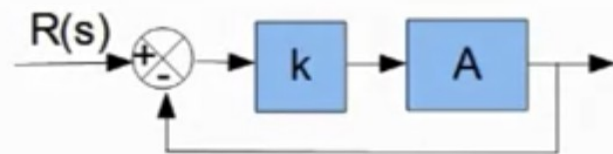
$$Y_R(s) = \frac{k(A + B)}{1 + k(A + B)} R(s)$$



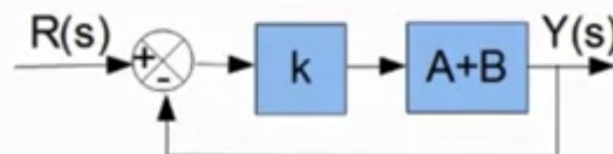
$$Y_N(s) = k A R(s)$$



$$Y_R(s) = k (A + B) R(s)$$



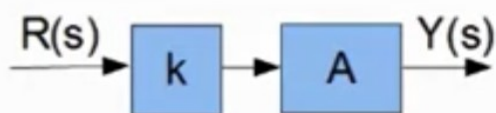
$$Y_N(s) = \frac{k A}{1 + k A} R(s)$$



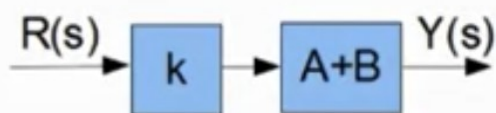
$$Y_R(s) = \frac{k (A + B)}{1 + k (A + B)} R(s)$$

$$Y_R(s) - Y_N(s) = k (A + B) R(s) - k A R(s)$$

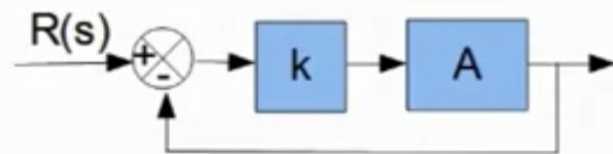
$$Y_R(s) - Y_N(s) = k B R(s)$$



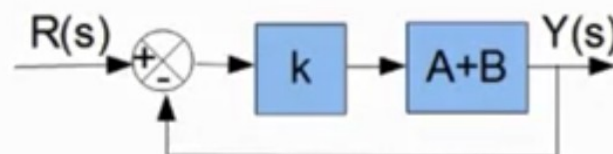
$$Y_N(s) = k A R(s)$$



$$Y_R(s) = k (A + B) R(s)$$



$$Y_N(s) = \frac{k A}{1 + k A} R(s)$$



$$Y_R(s) = \frac{k (A + B)}{1 + k (A + B)} R(s)$$

$$Y_R(s) - Y_N(s) = \frac{k (A + B)}{1 + k (A + B)} R(s) - \frac{k A}{1 + k A} R(s)$$

$$Y_R(s) - Y_N(s) = \frac{k B}{(1 + k (A + B))(1 + k A)} R(s)$$

Malha aberta: $Y_R(s) - Y_N(s) = k B R(s)$

$$Y_R(s) = k(A + B)R(s)$$

$$\frac{Y_R(s) - Y_N(s)}{Y_R(s)} = \frac{B}{A + B}$$

Malha fechada:

$$Y_R(s) - Y_N(s) = \frac{kB}{(1 + k(A + B))(1 + kA)} R(s)$$

$$Y_R(s) = \frac{k(A + B)}{1 + k(A + B)} R(s)$$

$$\frac{Y_R(s) - Y_N(s)}{Y_R(s)} = \frac{B}{(1 + kA)(A + B)}$$

Malha aberta:

$$\frac{Y_R(s) - Y_N(s)}{Y_R(s)} = \frac{B}{A + B}$$

Malha fechada:

$$\frac{Y_R(s) - Y_N(s)}{Y_R(s)} = \frac{1}{1 + kA} \frac{B}{A + B}$$

Malha aberta:

$$\frac{Y_R(s) - Y_N(s)}{Y_R(s)} = \frac{0,2 A}{1,2 A}$$

$$\frac{Y_R(s) - Y_N(s)}{Y_R(s)} = \frac{0,2}{1,2} = 0,17$$

Malha fechada:

$$\frac{Y_R(s) - Y_N(s)}{Y_R(s)} = \frac{1}{1 + k A} \frac{0,2 A}{1,2 A}$$

$$\frac{Y_R(s) - Y_N(s)}{Y_R(s)} = \frac{0,17}{1 + 10} = 0,015$$

Malha aberta:

20% de erro no modelo



17% de variação na saída

$$\frac{Y_R(s) - Y_N(s)}{Y_R(s)} = \frac{0,2 A}{1,2 A}$$

$$\frac{Y_R(s) - Y_N(s)}{Y_R(s)} = \frac{0,2}{1,2} = 0,17$$

Malha fechada:

20% de erro no modelo



1,5% de variação na saída

$$\frac{Y_R(s) - Y_N(s)}{Y_R(s)} = \frac{1}{1 + k A} \frac{0,2 A}{1,2 A}$$

$$\frac{Y_R(s) - Y_N(s)}{Y_R(s)} = \frac{0,17}{1 + 10} = 0,015$$

Malha aberta:

20% de erro no modelo



17% de variação na saída

$$\frac{Y_R(s) - Y_N(s)}{Y_R(s)} = \frac{0,2 A}{1,2 A}$$

$$\frac{Y_R(s) - Y_N(s)}{Y_R(s)} = \frac{0,2}{1,2} = 0,17$$

Malha fechada:

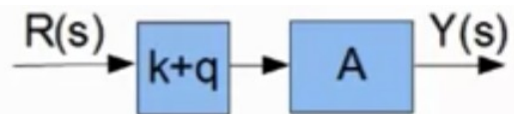
20% de erro no modelo



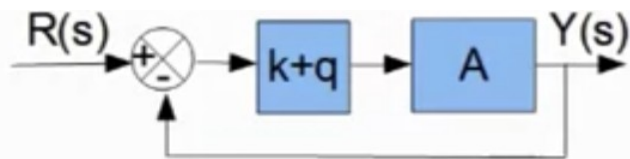
5,6% de variação na saída

$$\frac{Y_R(s) - Y_N(s)}{Y_R(s)} = \frac{1}{1 + k A} \frac{0,2 A}{1,2 A}$$

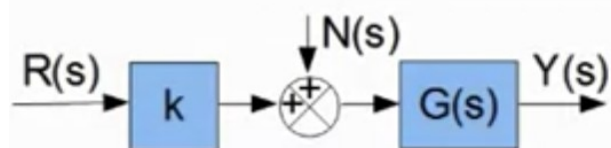
$$\frac{Y_R(s) - Y_N(s)}{Y_R(s)} = \frac{0,17}{1 + 2} = 0,056$$



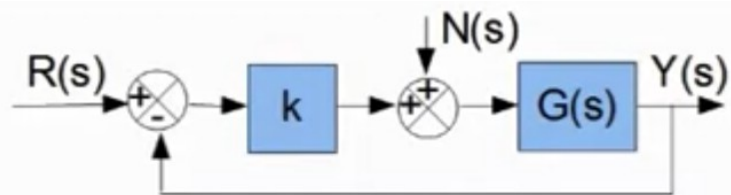
$$Y_R(s) = (k + q) A R(s)$$



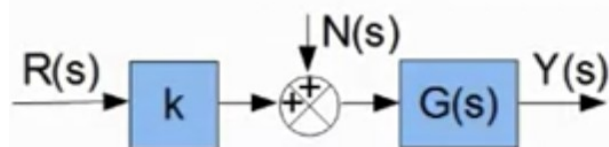
$$Y_R(s) = \frac{(k + q) A}{1 + (k + q) A} R(s)$$



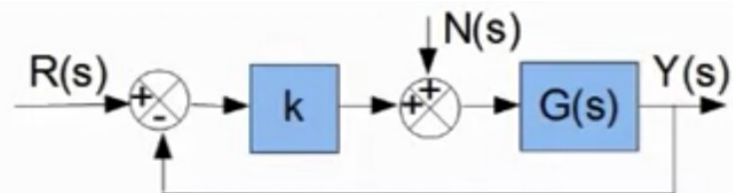
$$Y_N(s) = k A R(s)$$



$$Y_N(s) = \frac{k A}{1 + k A} R(s)$$



$$Y_R(s) = k A R(s) + A N(s)$$



$$E(s) = R(s) - Y(s)$$

$$U(s) = k E(s) + N(s)$$

$$U(s) = k R(s) - k Y(s) + N(s)$$

$$Y(s) = A U(s)$$

$$Y(s) = k A R(s) - k A Y(s) + A N(s)$$

$$Y_R(s) = \frac{k A}{1 + k A} R(s) + \frac{A}{1 + k A} N(s)$$

Malha aberta: $Y_N(s) = k A R(s)$

$$Y_R(s) = k A R(s) + A N(s)$$

$$Y_R(s) - Y_N(s) = k A R(s) + A N(s) - k A R(s)$$

$$Y_R(s) - Y_N(s) = A N(s)$$

Malha fechada:

$$Y_N(s) = \frac{k A}{1 + k A} R(s)$$

$$Y_R(s) = \frac{k A}{1 + k A} R(s) + \frac{A}{1 + k A} N(s)$$

$$Y_R(s) = \frac{k A}{1 + k A} R(s) + \frac{A}{1 + k A} N(s)$$

$$Y_R(s) - Y_N(s) = \frac{1}{1 + k A} A N(s)$$

Vantagens da realimentação:

- estabilizar o sistema;
- reduzir efeitos de erros;
- reduzir o efeito de perturbações.

Desvantagens:

- maior custo;
- maior complexidade;
- possibilidade de instabilizar um sistema estável.